



CLASS GUIDELINE

DNV-CG-0508

Edition August 2021

Smart vessel

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FOREWORD

DNV class guidelines contain methods, technical requirements, principles and acceptance criteria related to classed objects as referred to from the rules.

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CHANGES – CURRENT

This document supersedes the November 2020 edition of DNVGL-CG-0508.
The numbering and/or title of items containing changes is highlighted in red.

Changes August 2021

<i>Topic</i>	<i>Reference</i>	<i>Description</i>
Rebranding to DNV	All	This document has been revised due to the rebranding of DNV GL to DNV. The following have been updated: the company name, material and certificate designations, and references to other documents in the DNV portfolio. Some of the documents referred to may not yet have been rebranded. If so, please see the relevant DNV GL document. No technical content has been changed.

Editorial corrections

In addition to the above stated changes, editorial corrections may have been made.

CONTENTS

Changes – current.....	3
Section 1 General.....	6
1 Introduction.....	6
2 Objective.....	6
3 Scope.....	6
4 Application.....	6
5 References.....	6
6 Definitions and abbreviations.....	8
Section 2 Classification services.....	12
1 General.....	12
2 Additional class notation Smart.....	12
Section 3 Smart vessel and sustainable operation.....	14
1 General.....	14
2 Sustainable operation.....	14
3 Data management.....	19
Section 4 Enhancements.....	20
1 General.....	20
2 Operations and maintenance - machinery, equipment and systems (OPM).....	20
3 Operation and maintenance - hull and structure (OPH).....	24
4 Energy efficiency and environment (EEN).....	25
Section 5 System qualification.....	27
1 General.....	27
2 System qualification process.....	27
3 Implementation and testing.....	28
Section 6 Models and applications.....	29
1 General.....	29
2 Verification of models and data-driven applications.....	30
3 Development.....	30
4 Model performance.....	31
5 Confidence level.....	32
6 Limitations and operating conditions.....	32

Appendix A Operations and maintenance - machinery, equipment and systems...	33
1 Introduction.....	33
2 Enhancements.....	33
Appendix B Operation and maintenance - hull and structure.....	38
1 Introduction.....	38
2 Enhancements.....	38
Appendix C Energy efficiency and environment.....	41
1 Introduction.....	41
2 Additional notations.....	41
3 Performance monitoring solutions.....	44
Changes – historic.....	46

SECTION 1 GENERAL

1 Introduction

Technology development with widespread use of digital solutions and increased connectivity available for the maritime industry is enabling use of data and information to further optimize vessels' operations and reduce the environmental footprint. Efficient scale-up requires reliable data collection and exchange of data that will also require a certain level of standardization and data quality management. Digitalization also introduces new risks towards cyber security and data security which should be managed.

2 Objective

The objective of this class guideline (CG) is to provide guidance for the additional class notation **Smart**. The CG also provides a method for a system qualification (SQ) process and verification of digital and automated reporting tools.

3 Scope

This CG provides additional information on the requirements given in the **Smart** notation and shows how digitalization and innovative solutions may be utilized to facilitate effective and sustainable operation and management of a vessel.

4 Application

This CG provides those involved in operating and maintaining a vessel a description of how an organization can use the **Smart** notation as an integrated part of their asset management in order to improve performance and achieve a sustainable operation.

The CG also provides those involved in planning and designing a vessel relevant information that can be used to develop their products and services to improve operation and maintenance using new technology and data from sensors and other sources of information as a basis for decisions and improved monitoring and control.

This CG can be used as guidance when referred to by the rules.

The rules described in [DNV-RU-SHIP Pt.6 Ch.5 Sec.24](#) provide the requirements and have precedence over this CG.

5 References

5.1 DNV documents

[Table 1](#) lists DNV references used in this document.

Table 1 DNV documents

<i>Document code</i>	<i>Title</i>
DNV-CG-0264	Autonomous and remotely operated ships
DNV-CG-0557	Data-driven verification
DNV-CG-0564	Data collection infrastructures
DNV-CP-0484	Approval of service supplier scheme
DNV-CP-0500	Automated boiler water maintenance system

<i>Document code</i>	<i>Title</i>
DNV-CP-0569	Electronic record book - MARPOL
DNV-RP-0496	Cyber security resilience management for ships and mobile offshore units in operation
DNV-RP-0497	Data quality assessment framework
DNV-RP-0510	Framework for assurance of data-driven algorithms and models
DNV-RU-OU-0300	Fleet in service (MOU)
DNV-RU-SHIP Pt.6 Ch.5 Sec.24	Smart vessel - Smart
DNV-RU-SHIP Pt.6 Ch.11	Digital features
DNV-RU-SHIP Pt.7 Ch.1	Survey requirements for fleet in service

5.2 External documents

Table 2 lists external references used in this document.

Table 2 External documents

<i>Document code</i>	<i>Title</i>
EN 13306	Maintenance - Maintenance terminology
IEC 61162	Maritime navigation and radiocommunication equipment and systems - Digital interfaces
ISO 5725-1	Accuracy (trueness and precision) of measurement methods and results — Part 1: General principles and definitions
ISO 8000-2	Data quality - Part 2: Vocabulary
ISO 9001	Quality management systems - Requirements
ISO 12482	Cranes — Monitoring for crane design working period
ISO 13372	Condition monitoring and diagnostics of machines - Vocabulary
ISO 13379-1	Condition monitoring and diagnostics of machines- Data interpretations and diagnostics techniques - Part 1: General guidelines
ISO 13381-1	Condition monitoring and diagnostics of machines - Prognostics - Part 1: General guidelines
ISO 15663-1	Petroleum and natural gas industries — Life cycle costing — Part 1: Methodology
ISO 17359	Condition monitoring and diagnostics of machines - General guidelines
ISO 18436	Condition monitoring and diagnostics of machines — Requirements for qualification and assessment of personnel
ISO 21745	Electronic record books for ships - technical specification and operational requirements
ISO 55000	Asset management - Overview, principles and terminology

6 Definitions and abbreviations

6.1 Definition of verbal forms

The verbal forms defined in [Table 3](#) are used in this document.

Table 3 Definition of verbal forms

<i>Term</i>	<i>Description</i>
shall	verbal form used to indicate requirements strictly to be followed in order to conform to the document
should	verbal form used to indicate that among several possibilities one is recommended as particularly suitable, without mentioning or excluding others
may	verbal form used to indicate a course of action permissible within the limits of the document

6.2 Definition of terms

The terms defined in [Table 4](#) are used in this document.

Table 4 Definition of terms

<i>Term</i>	<i>Definition</i>
accuracy	for measurements, the combination of trueness and precision, see ISO 5725-1
availability	ability of an item to be in a state to perform as and when required, under given conditions, assuming that the necessary external resources are provided, see EN 13306
condition monitoring	activity, performed either manually or automatically, intended to measure at predetermined intervals the characteristics and parameters of the actual state of the item, see EN 13306
condition based maintenance	preventive maintenance which include a combination of condition monitoring and/or inspection and/or testing, analysis and the ensuing maintenance actions, see EN 13306
confidence level	figure of merit (e.g. percentage) that indicates the degree of certainty that the prediction (diagnosis/prognosis) is correct, see ISO 13381-1
cyber security	practices, tools and concepts that protect: — the operational technology (OT) against the unintended consequences of a cyber incident — information and communications systems and the information contained therein from damage, unauthorised use or modification, or exploitation — against interception of information when communicating and using the internet See DNV-RP-0496 .
data	reinterpretable representation of information in a formalized manner suitable for communication, interpretation, or processing, see ISO 8000-2 In this context data refers to time series data, event data (including IEC 61162 sentences), time series meta data and information models.
data collection infrastructure	hardware and software which provide end to end communication of data from a vessel to a remote data server or a specific data consumer, see DNV-CG-0564

<i>Term</i>	<i>Definition</i>
data-driven application	application which includes at least one data-driven model. The parameters and configuration of a data-driven model are automatically determined (or learned) from data using a suitable algorithm, see DNV-RP-0510 .
data-driven model	simple trending, neural network, pattern recognition/correlations, statistical data analysis, etc.
data quality management	coordinated activities to direct and control an organization with regards to data quality, see ISO 8000-2
data quality	measurement to which degree data meet the implicit or explicit expectations and requirements of users or systems utilizing the data. Information and data quality are defined and measured according to syntactic, semantic and pragmatic quality. Syntactic and semantic quality is measured through a verification process, whereas pragmatic quality is measured through a validation process, see DNV-RP-0497 .
dependability	ability to perform as and when required, see EN 13306
descriptor	data item derived from raw or processed parameters or external observation
diagnostics	examination of symptoms and syndromes to determine the nature of faults or failures (kind, situation, extent), see ISO 13372
enhancement	class services like notations, survey arrangements and other technical solutions used to achieve enhanced performance or service for the vessel. Enhancements are being applied as elements to obtain one of the qualifiers for the Smart notation. Enhancements may be applicable for a ship and MOU, or both.
failure mechanism	physical, chemical or other processes which may lead or have led to failure, see EN 13306
failure mode	manner in which the inability of an item to perform a required function occurs, see EN 13306
maintenance	combination of all technical, administrative and managerial actions during the life cycle of an item intended to retain it in, or restore it to, a state in which it can perform the required function, see EN 13306
maintenance management	all activities of the management to determine the maintenance requirements, objectives, strategies and responsibilities, and implementation of them by such means as maintenance planning, maintenance control, and the improvement of maintenance activities and economics, see EN 13306
maintenance objectives	targets assigned and accepted for the maintenance objectives
maintenance plan	structured and documented set of tasks that include activities, procedures, resources and the time scale required to carry out maintenance, see EN 13306
maintenance strategy	management method used to achieve the maintenance objectives, see EN 13306
maintenance support	provisions of resources, services and management necessary to carry out maintenance, see EN 13306
model	representation of some process and/or entity, which is typically used to make predictions and/or other useful transformations of input data
novel technology	technology, products or systems for which complete performance or approval criteria does not exist, see DNV-CG-0264

<i>Term</i>	<i>Definition</i>
organization	person or group of people that has its own functions with responsibilities, authorities and relationships to achieve its objectives, see ISO 9000 In this context an organization refers to a vessel owner or management assigned by the owner.
parameter	variable representing some significant measurable system characteristic, see ISO 13372
predetermined maintenance	preventive maintenance carried out in accordance with established intervals of time or number of units of use but without previous condition investigation, see EN 13306
preventive maintenance	maintenance carried out at predetermined intervals or according to prescribed criteria and intended to reduce the probability of failure or the degradation of the functioning of an item, see EN 13306
predictive maintenance	condition based maintenance carried out following a forecast derived from repeated analysis or known characteristics and evaluated of the significant parameters of the degradation of the item, see EN 13306
precision	closeness of agreement between independent test results obtained under stipulated conditions, see ISO 5725-1
prognostics	analysis of the symptoms of faults to predict future condition and residual life within design parameters, see ISO 13372
physics-based model	first principle equations, finite element methods, etc.
reliability	ability of an item to perform a required function under given conditions for a given time interval, see EN 13306
strategy	general plan to achieve one or more long-term or overall goals under conditions of uncertainty. A maintenance strategy is a management method used in order to achieve maintenance objectives, see EN 13306
sustainability	ability to exist constantly
symptom	perception, made by means of human observations and measurements, which may indicate the presence of one or more faults, see ISO 13372
syndrome	group of signs or symptoms that collectively indicate or characterize an abnormal condition, see ISO 13372
trueness	closeness of agreement between the average value obtained from a large series of test results and an accepted reference value, see ISO 5725-1

6.3 Abbreviations

The abbreviations described in [Table 5](#) are used in this document.

Table 5 Abbreviations

<i>Abbreviation</i>	<i>Description</i>
ASP	alternative survey programme
CBM	condition based maintenance
CG	class guideline

<i>Abbreviation</i>	<i>Description</i>
CM	condition monitoring
CMMS	computerized maintenance management system
CMON	construction monitoring
CoC	continuous certification services
DCBM	drilling equipment condition based maintenance
DDV	data-driven verification
EEN	energy efficiency and environment enhancement
EES	electrical energy storage
ERB	electronic recording book
ERP	enterprise resource planning (tool)
ESIM	electrical system integrity management
ESV	enhanced system verification
FMSA	failure modes symptoms analysis
FMS	fatigue methodology specification
INS	integrated navigation system
ISDS	integrated software dependent systems
MaaS	machine-as-a-service
MARPOL	international convention for the prevention of pollution from ships
MMC	machinery maintenance connect
MOU	mobile offshore unit
MPMS	machinery maintenance system
OPEX	operational expenditure
OPH	operational and maintenance enhancement - machinery, equipment and systems
OPM	operational and maintenance enhancement - hull and structure
PMS	planned maintenance system
RBI	risk based inspection
RCM	reliability centred maintenance
REW	remote witnessing
RIT	remote inspection techniques
SaaS	software-as-a-service
SQ	system qualification
TOC	total cost of ownership

SECTION 2 CLASSIFICATION SERVICES

1 General

1.1 Class notation

Class notations are assigned to a vessel in order to determine applicable rule requirements for assignment and retention of class. Class notations are either mandatory or optional. For more details about class notations, see [DNV-RU-SHIP Pt.1 Ch.2](#).

1.2 Survey arrangement

A survey arrangement is a systematic approach agreed between an owner/manager of a vessel and the Society, defining how and when hull and structure, equipment and/or systems are surveyed.

Guidance note:

A survey arrangement may include additional requirements for the vessel management and/or the use of an approved service supplier, see [\[1.3\]](#).

Systems used to support a survey arrangement may be approved at newbuilding stage, while any initial survey that is required for some survey arrangement's should be done after the system has been in operation onboard for six (6) months or more.

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1.3 Approved service supplier

An approved service supplier is a company providing services to the customer, such as measurements, tests or maintenance of safety systems and equipment, the result of which may form the basis for the Society's decisions. Some survey arrangements have a specific requirement of using an approved service supplier covering verification requirements for inspection, maintenance and/or testing as described in the rules for the different survey arrangement.

Guidance note:

For more information on approval of service suppliers, see [DNV-CP-0484](#).

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1.4 Performance monitoring solution

Performance monitoring solutions are processes and means introduced to better understand different aspects of a vessel's or a fleet of vessels' performance. Performance monitoring may be applied on different areas, and is often monitored towards set performance indicators. In the context of the **Smart** framework, performance monitoring normally refers to energy efficiency and environmental aspects.

2 Additional class notation Smart

2.1 General

The additional class notation **Smart** promotes and visualizes a shift towards more efficient and green operations that are based on digital or environmentally-friendly solutions, improving the performance and operation of a vessel.

2.2 Smart framework

Smart provides a framework for profiling technical solutions and class services that utilize digital and technical solutions to improve the operational efficiency or environmental footprint of the vessel. These technical solutions and class services are denoted as enhancements and used as a collective term. Enhancements are additional class notations and survey arrangements that represent digital capabilities, enhanced vessel performance and promote energy efficiency and environmental friendly solutions. Enhancements can also be novel technology and solutions approved by the Society following a system qualification (SQ) process, see [Sec.5](#). See also [Sec.4](#) for more information on enhancements.

The notation has three (3) qualifiers, each referring to a range of available enhancements that may be selected to obtain the applicable qualifier. A vessel shall obtain minimum one (1) qualifier in order to be granted the additional notation **Smart**.

The framework is illustrated in [Figure 1](#).

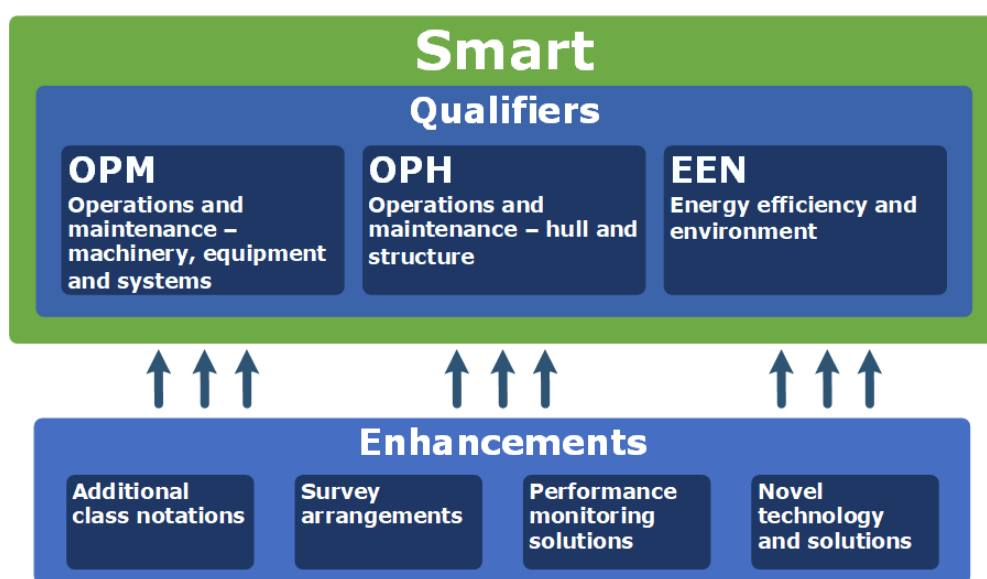


Figure 1 Smart framework

SECTION 3 SMART VESSEL AND SUSTAINABLE OPERATION

1 General

This chapter provides guidance and recommendations regarding the application of the **Smart** framework. It provides examples and illustrates how an organization may use the framework as an integral part of asset management in order to optimize performance and achieve a sustainable operation.

2 Sustainable operation

2.1 General

The maritime industry is transitioning like most other industries towards digitalization, enabling new opportunities by using data streams from sensors and other sources of information as basis for decisions and enhanced monitoring and control.

The industry is also faced with an increasing pressure from regulators, governments and end customers to reduce harmful emissions with outlined strategies to realise zero-emission from shipping by the middle of the century.

Change becomes no longer an option but a necessity for survival and success. This requires organizations to re-think their current strategies and change and develop to achieve a sustainable operation that can strengthen their competitiveness in both the short term and the long term.

Smart is a term based on a set of fundamentals comprising digitalization and new technology, energy efficient and green operation, and innovative and sustainable solutions. In this context digitalization becomes an important part of the continuous process to establish effective solutions and systems to improve organizational knowledge and decision making, enhance vessel performance and reduce operational cost (OPEX). A smart vessel in this context is consequently not only focused on technology and solutions enabling digital capabilities. The main objective of a smart vessel should be a sustainable operation by applying relevant and available solutions to optimize performance throughout the life cycle of the vessel. **Smart** is thus in addition to digital solutions and new technology pointing to solutions supporting energy efficiency and green operations.

Optimizing the life cycle of the vessel should begin at conceptual design and continue through construction, commissioning and the operations phase. Life cycle management (LCM) focuses on design and solutions that are especially selected and adapted according to vessel operation and maintenance, performance requirements, management support and relevant stakeholders.

Guidance note:

For more information on life cycle management and life cycle costing, see ISO 15663-1.

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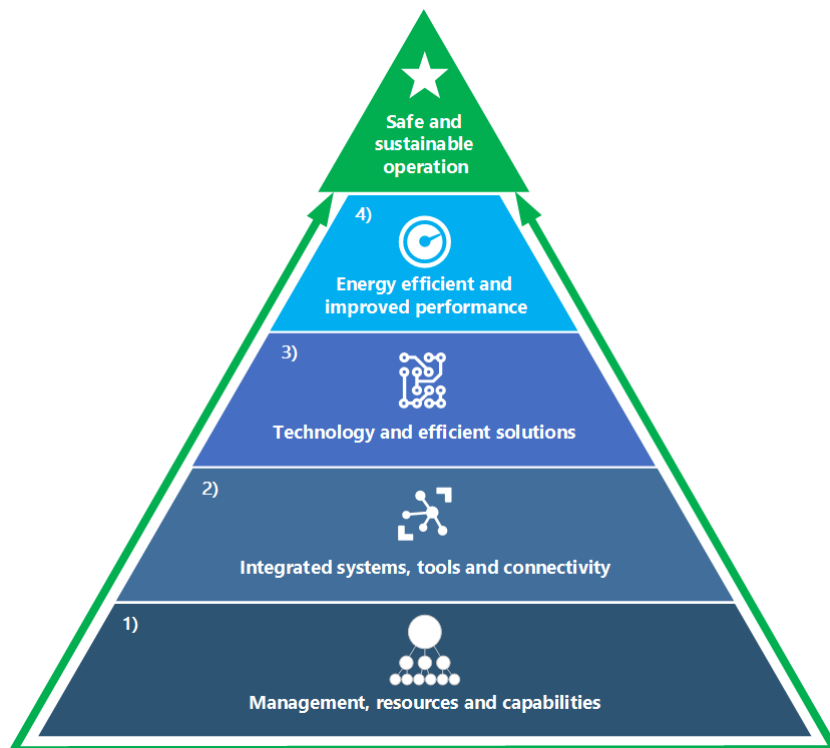


Figure 1 Sustainable operations

Figure 1 illustrates how important elements should be structured and connected to establish an organization with the right type of management, resources, systems and solutions in order to obtain sustainable operations.

Guidance note:

The effects of implementing new technology and solutions may in many cases be limited if the organization does not have the necessary capabilities and support systems in place to utilize the full potential provided by the new technology and solutions. Some organizations will in many cases only focus on the technology and solutions, without performing the basic part of adapting the organization to fully support the new technology and solutions. This is also why these elements are emphasised and described in more details in this CG.

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2.2 Management, resources and competence

2.2.1 General

The first element in Figure 1 addresses the organizational structure, management, resources and capabilities.

2.2.2 Asset management

Asset management is the system and continued process enabling an organization to realize value from assets (vessel) through achieving the organization's objectives using strategies to strengthen their competitiveness and operational sustainability.

An asset management system is established based on:

- leadership
- planning

- support
- operation
- performance evaluation
- improvement.

These elements describe a system and continued process that can be applied to all parts of the organization working in a standardized and integrated approach.

Guidance note:

For more information on asset management, see ISO 55000.

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2.2.3 Objectives and strategy for a smart vessel

Asset management objectives and strategy should be developed based on a good understanding of the industry's current state, where the organization wants to go, what the organization needs and identifying any gaps that should be closed.

Guidance note:

Objectives and strategy should be revised and updated in accordance with experience and development in the industry as part of a continuous improvement process.

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For a smart vessel and the organization, the objectives may include the fundamentals of **Smart**:

- digitalization and new technology
- energy efficient and greener operations
- innovative and sustainable solutions.

To achieve the objectives, a suitable strategy should be selected that will apply to any part of the organization. The strategy should be based on elements that promote the objectives and represent focus areas for the organization of how they should achieve the objectives today and in the future.

Guidance note:

Strategy elements for a smart vessel and the organization should be selected based on the asset management objectives promoting e.g. increased:

- knowledge and decision-making using data and analytics
- process efficiency and automation
- centralization of processes and support
- utilization of vessel and resources
- standardization.

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Based on the organization's asset management objective and strategy, relevant parts of the organization should develop their specific objectives and strategy adapted to their application and/or service.

Guidance note:

Relevant parts of the organization may be operations, maintenance, projects/newbuilding, IT, procurements, etc.

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2.2.4 Action plan

In order to execute the specific objectives and strategy, each part of the organization should develop an action plan detailing how to achieve the intended outcome according to the specific strategy elements. The action plan should include specific solutions and systematics, and also address development and implementation, resources, tools and technology. The action plan should include a time frame for each solution including development and implementation. An action plan should not only address short term solutions but also include a plan for more long term solutions that represents future developments and improvements to be implemented when the organization has gained knowledge and experience. This will provide guidance and direction for the organization working towards a defined and common goal. It will also

provide a better understanding of what is needed in terms of development across the different parts of the organization to achieve the intended outcome.

Selection of tools and solutions applicable for the vessel should be based on an evaluation of:

- type of vessel
- vessel management organization, its structure and resources
- fleet and vessels under same management
- objectives, strategy and plan, see [2.2.3]
- cost vs. benefit (analysis).

This type of evaluation and analysis may help determine the feasibility and how well applicable solutions comply with the specific vessel and its management. It may also help determine if the initial investment related to the solution may be distributed by implementing the solution on a number of vessels under same or similar management. A simple cost vs. benefit analysis may be used to measure the effectiveness of the investment compared to the current or traditional approach.

2.2.5 Collaboration and cooperation

With digitalization also comes the increase of complexity that calls for new knowledge, technology and systems. In many cases this requires changes to the organization including both management and other resources. In some cases more collaboration and cooperation among industry actors and stakeholders such as yard, OEM, suppliers and other specialist companies should be considered to utilize the use of resources.

Guidance note:

As with many other industries, the maritime industry see a shift towards cooperation and finding new ways of working. This can relate to services where the organization does not have the sufficient capability and/or resources internally to provide an optimal service or solution. It can also relate to purchase and service agreements where OEM and other suppliers are offering different alternatives from a standard purchase cost agreement, that normally also includes a large amount of spare parts, to alternatives like:

- purchase + maintenance services (predetermined maintenance, CM and CBM)
- purchase + maintenance and continuous certification services (CoC)
- purchase + total cost of ownership (ToC) where the equipment owner pays for the usage of the equipment (fixed fee)
- machine-as-a-service (MaaS) where the equipment owner pays for the usage typically with no initial purchase cost
- software-as-a-service (SaaS) where companies provides a complete software solution (software, cloud, platform, etc.) that a client purchases on a pay-as-you-go basis.

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2.3 Integrated systems, tools and connectivity

The second element addresses how the increase in connectivity and digital work flows will introduce the need for data platform(s), new tools and the integration between existing and new systems enabling sharing of data and usage across different systems. This also introduces the need for standardization and capabilities for scaling in accordance with technology development and requirements from the maritime industry of tomorrow. An important aspect when selecting and implementing new systems and tools is the integration with work processes and the people using them.

When selecting tools and systems it should also be considered how common systems can be used across different parts of the organization to limit the number of similar systems that will drive the cost for maintenance and support.

Guidance note:

A basic example of a more common system that may be used across the organization may be illustrated by an enterprise resource planning (ERP) tool compared to a computerized maintenance management system (CMMS). The CMMS is usually limited to maintenance management and used to monitor and record all maintenance work and resources, while the ERP system is a software package with integrated applications that work together to manage and automate all parts of the organization including operations and maintenance, human resources, accounting and financials, inventory, procurement and project management. The ERP system can be used to collect, store, manage, and interpret data from many business activities and can have the capacity to integrate with other systems like a data collection infrastructure, see [DNV-RU-SHIP Pt.6 Ch.11 Sec.1](#). The ERP system is more complex, requires more support and usually involves a higher initial cost. The benefit comes by utilising the capabilities of the system across the organization and the cost can be significantly less seen over time and over the life cycle of the vessel(s).

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2.4 Technology and efficient solutions

The third element represents technology and solutions that should be integrated with the support systems and focus on providing energy efficient and improved performance to reduce the environmental impact as illustrated in the fourth element in [Figure 1](#).

Smart represents a variety of solutions denoted as enhancements, see [Sec.4](#). These are solutions that are based on already developed notations or survey arrangements, or solutions not covered by existing rules, qualified on a case-by-case basis, see [Sec.5](#). Solutions not covered by the rules are an important part of the notation as the qualification of new technology and solutions may contribute to development and innovation that is one of the fundamental elements in **Smart**.

Some enhancements will normally require an initial investment of resources (equipment, tools, systems, development, implementation and support). By applying the same enhancement for a group or fleet of vessels the cost may be distributed and reduced per vessel. Enhancements may have same or similar requirements for resources and then further provide similar benefits by combining the selection of enhancements initially or over a time period according to a development plan.

Selection of enhancements should consider the organization's objectives and strategy, see [\[2.2.3\]](#) to ensure the selection of solutions, sometimes requiring new technology, tools and resources, to be in line with the common direction set for the organization. This may also benefit the organization to better understand the need for solutions and investments across different parts of the organization resulting in more common solutions reducing the total cost over time.

Selection of enhancements and solutions should be based on the life cycle of the vessel, and as much as possible account for predicted trends in the market and outlined strategies of applicable regulatory regimes to avoid costly technology retrofits in the future.

2.5 Energy efficient and improved performance

The fourth element illustrates how the use of technology and solutions which are cost, energy and emission efficient will provide results to the vessel performance. This element also reflects the results from an optimized organization that utilizes available resources and leverages digital technologies in their ways of working.

Performance monitoring should be used to measure and evaluate the vessel performance. The monitoring may be used for measuring the effectiveness of technology and solutions, improvements or internal and external benchmarking, providing regulators and governments with transparency and enabling more efficient ways of demonstrating compliance. The **Smart** notation represents a selection of performance monitoring solutions that may be used for compliance. This type of solution typically sorts under the **EEN** qualifier, see [Sec.4 \[4\]](#).

2.6 Sustainable operation

An organization having a sustainable operation may be characterized by a well defined company objective and strategy, see [\[2.2.3\]](#), a strong ability to compete in the markets of today and tomorrow, and an

operation that has a clean environmental profile including solutions which provide measurable reductions in emissions and impact on the environment.

3 Data management

Most organizations want to leverage digital technologies and utilize the potential of their data and information collected from sensors and source systems used as basis for decisions and input for technology and solutions. With a rapidly increasing amount of data and data based solutions integrated as part of the vessel operation, this introduces a risk of using insufficient or incomplete data, leading to potentially inaccurate predictions and sometimes wrong decisions. Using data as an integrated part of the vessel operations will require an active data quality assurance to monitor and correct the quality of data. The increase in connectivity and integration of systems also introduces risk for cyber security threats that need to be prevented, mitigated and responded to.

Data management is a term used collectively to describe the combination of technical solutions including network, hardware, software and a management (organization and resources) with coordinated activities ensuring sufficient:

- data collection infrastructure
- data quality monitoring
- data security (cyber security).

Rules for data management are described in the rules for **D-INF**, see [DNV-RU-SHIP Pt.6 Ch.11 Sec.1](#).

Detailed requirements for certification of data collection infrastructures are described in [DNV-CG-0564](#).

SECTION 4 ENHANCEMENTS

1 General

Enhancements are notations, survey arrangements, and other technical solutions used in combination to obtain applicable qualifiers.

Enhancements are grouped into three (3) different subjects each representing a qualifier for **Smart**:

- operations and maintenance - machinery, equipment and systems (OPM), see [2]
- operations and maintenance - hull and structure (OPH), see [3]
- energy efficiency and environment (EEN), see [4].

Guidance note:

In some cases an enhancement can be categorized under a specific group even if it partly complies with or could be sorted under another group. Some enhancements may also be used to obtain more than one qualifier.

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Operations and maintenance is the combination of process, administration, resources, tools and systems needed to ensure that the vessel, equipment and systems are able to perform as and when required (dependability). Operations and maintenance are used in combination because a vessel cannot operate at maximum efficiency without being sufficiently maintained.

The enhancements within each qualifier are selected solutions applied to the vessel that may enhance the performance and operation in a life cycle context by optimizing the design, use alternative technical solutions to improve performance and utilize data for analysis and applications.

Enhancements can be:

- 1) additional class notations
- 2) survey arrangements
- 3) performance monitoring solutions
- 4) novel technology and solutions, see [Sec.5](#).

Guidance note:

Novel technology may be based on (build on) additional notations or established as a new solution if the Society does not have applicable and existing rules and/or regulations.

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The class societies' main focus is to ensure the safety and integrity of the vessel and operation. The selection of enhancements reflects this focus, but operations and maintenance enhancements also include services that can benefit operational performance including e.g. solutions providing better performance in terms of fuel consumption, sailing time, cargo handling and reduced off-hire and time spent in port.

Operational performance is also closely linked to the selection of energy efficiency and environmental enhancements that focus on cost-, energy- and emission-efficient operation of the vessel.

2 Operations and maintenance - machinery, equipment and systems (OPM)

2.1 General

Operations and maintenance optimization is an important contribution for a sustainable operation and to reduce OPEX. In this respect operational and maintenance enhancements are solutions that aim at increasing the efficiency of the operation by implementing well adapted management systems and tools to reduce cost and risk related to maintenance, inspection and testing activities, see also [3].

Machinery, equipment and systems applicable to main class, service notations and additional class notations will be subjected to periodical surveys and inspected, tested and maintained on a continuous basis to ensure

it remains in satisfactory condition and in compliance with verification requirements and accepted standards, see [DNV-RU-SHIP Pt.7 Ch.1 Sec.1](#).

The survey scope normally consists of examination, verification and testing as required for different surveys. The survey may expose the vessel to non-productive time associated with taking the vessel out of service to perform the applicable survey scope. The survey scope for a vessel is based on vessel type, service notation, additional notation(s) and applicable survey arrangement(s).

Operations and maintenance - machinery, equipment and systems (OPM) enhancements includes solutions that use data as an important element and provides options for maintenance management and solutions that may optimize inspection, testing and verification activities.

OPM enhancements may be used to obtain the **OPM** qualifier in **Smart**. For more information, see [App.A](#) for applicable OPM enhancements.

2.2 Maintenance management

2.2.1 General

All vessels will be subjected to a survey arrangement, see [Sec.2 \[1.2\]](#) applicable for machinery and equipment in accordance with requirements of the rules. The basic (default) requirement is that machinery and equipment are inspected every five (5) years. This may introduce unnecessary inspection activities and over-maintenance.

As an alternative, a vessel and organization may obtain a survey arrangement based on the following approaches:

- planned maintenance system (PMS): predetermined maintenance based on OEM recommendations
- condition based maintenance (CBM): predictive maintenance using an approved service supplier, see also [\[2.2.2\]](#)
- reliability based planned maintenance system (PMS RCM): preventive maintenance based on reliability centred maintenance (RCM) analysis.

Guidance note:

Survey arrangements for PMS will only qualify as an enhancement in **Smart** in combination with machinery maintenance connect (MMC). PMS is also considered a minimum basis for a maintenance management and a prerequisite and enabler for CBM.

PMS RCM is only applicable for MOU (offshore) and applies for machinery and equipment covered by main class and/or additional notations, see [DNV-RU-OU-0300 Ch.2 Sec.2 \[4\]](#). PMS RCM may also be combined with MMC.

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Survey arrangements offering an alternative approach may provide more flexibility and maintenance better adapted to the machinery and equipment's usage and/or condition, providing optimal utilization. See [App.A](#) for more information and survey arrangements that may be used for compliance for the **OPM** qualifier.

Survey arrangements should be selected and aligned with the applicable maintenance management plan for the vessel and organization. The maintenance management plan should address specific initiatives and solutions to be implemented which will contribute to the success of the maintenance management objectives and strategy. A maintenance management plan should include (but not limited to) the following elements:

- development process including plan and time frame
- internal resources (management, competence and responsibilities)
- outsourcing (service suppliers)
- tools and support systems (ERP, CMMS and data collection infrastructure)
- maintenance programme (development and/or optimization)
- quality assurance and control
- performance monitoring and continuous improvement.

Guidance note:

The maintenance programme should include all maintenance activities for an item(s) intended to retain it, or restore it, to a state in which it can perform the required function.

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2.2.2 Condition based maintenance

Damage and fault can occur due to special events, degradation mechanisms and poor design and workmanship.

Condition monitoring (CM) refers to a process of monitoring parameters (sensor data) representing a degradation typically related to a known failure mode in order to identify a change which is indicative of a developing fault or state (condition).

CM is predictive maintenance and a proactive approach that allows maintenance, inspection and testing to be scheduled to prevent consequential damages and avoid its consequences. CM has the benefit of providing early warning and detection of conditions that would shorten normal lifespan, which can be addressed before they develop into a failure. CM can be used on demand or based on continuous or periodic monitoring for in-service machinery and equipment and hull and structural components.

Maintenance of machinery and equipment may include CM as part of an applicable maintenance programme. CM should then be an integrated part of the maintenance programme and maintenance management for the vessel. CM can be used by applying methods of monitoring degradations and failure mechanisms related to specific failure modes. CM should only be used where predictive maintenance is regarded as an applicable and effective maintenance approach (cost vs. benefit).

CM for machinery and equipment can be based on measuring techniques (manipulated parameters), e.g. vibration analysis, ultrasound or other failure modelling techniques that may be knowledge-based, physics-based or data-driven. It may also be used as part of an integrated approach like condition-based maintenance (CBM) combining parameters and information acquired from testing and inspection (condition assessment) with measurement techniques and other failure modelling techniques. [Figure 1](#) illustrates the different elements part of CBM.

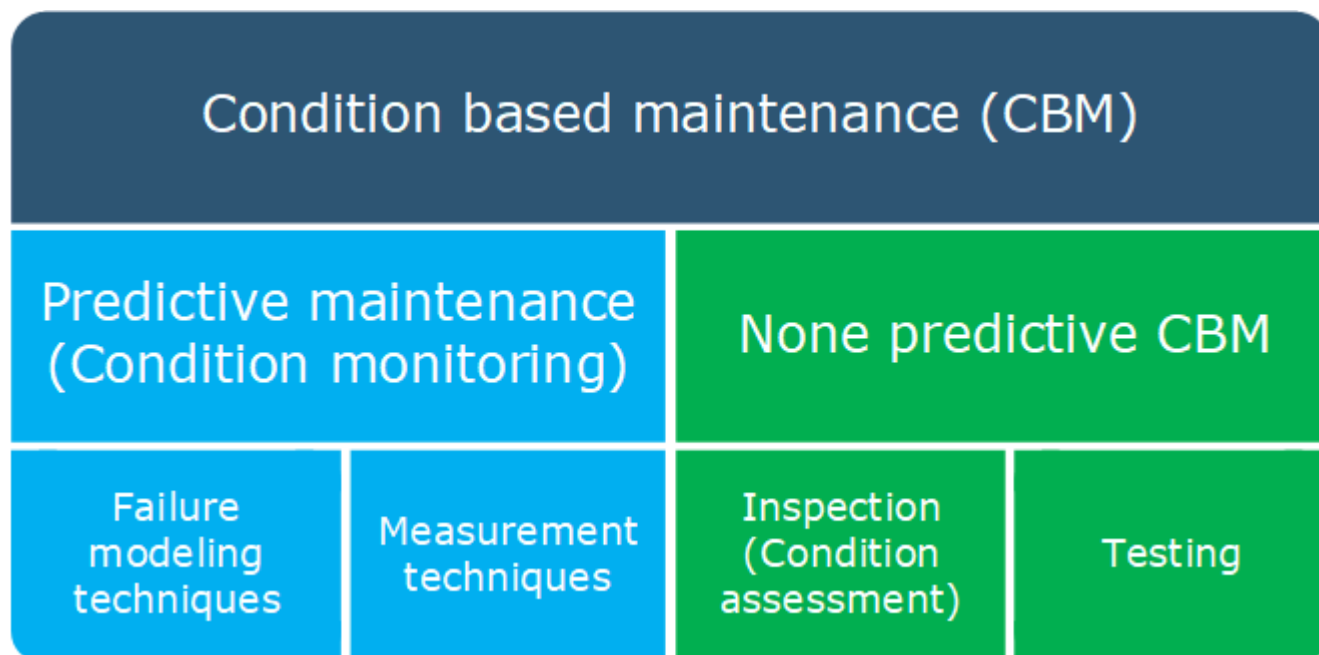


Figure 1 Condition based maintenance

If CM/CBM should be applied as part of a maintenance programme for machinery and equipment covered by main class or additional notations, it will normally require use of an approved service supplier, and the vessel should select an applicable survey arrangement according to application and type of vessel.

Guidance note:

For more information about survey arrangements for CM and CBM, see [App.A](#). For information regarding approval of service suppliers for CM and CBM services and other applicable programmes, see [DNV-CP-0484](#).

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2.3 Systems

Any control system, sub-systems and software components related to functions covered by main class and additional notations will be subject to verification and testing on a continuous basis. This usually involves a reduced scope annually and a more complex test scope every five (5) years. The test scope should ensure that the system performs according to the approval or agreed standard and that the equipment controlled by the system performs as intended.

Verification and testing activities can in many cases represent a significant disruption to the operation of the vessel leading to unproductive time due to the test activity and potential temporary unavailability of functions due to a failed test.

Verification and test activities may be optimized by introducing solutions and systematics that can distribute the test activity over the five (5) year period to avoid a major test scope every five (5) years. The test activity may also be adapted and scheduled according to operation to ensure minimum disruption.

Systems may also be subjected to additional test and verification methods that may be utilized to assist in verification of the functionality and performance of selected systems in order to provide objective evidence of acceptable functionality and quality according to stated requirements. The objective is to ensure system reliability and detect any fault or defect that can potentially cause a failure and ultimately loss of a function. The methods may also be used to detect any fault or defect introduced by changes or modifications made to the system.

The additional notation **DDV** is a concept providing alternative methods and systematics for verification of maritime and offshore cyber physical systems to be verified based on data-driven verification methods. The objective is to ensure that systems are designed to be verified by verify on-demand capabilities. The methods may upon request be accepted as additional to, or replace evidence generated from traditional verification activities, that traditionally have relied on deploying a surveyor to the vessel, as required by the applicable rules. The intention is that data-driven verification methods may be accepted if the new methods provide the same, or higher, level of assurance as the traditional methods. Application of data-driven verification of a system is subject to approval by the Society, unless specifically allowed in the rules governing the specific equipment.

By introducing alternative methods for verification and testing, a vessel may increase utilization by reducing exposure to non-productive time associated with taking the vessels out of service for trials, reducing costs associated with surveyors attending the vessel, reducing redundant verification activities, and reducing the cognitive burden imposed upon vessel personnel (crew) carrying out the testing.

Guidance note:

It is recommended to apply the systematics described for additional class notations like **DDV**, **ESV** and **ISDS** for verification and test methods, see [App.A](#) for more information.

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The inspection and test activity may also be optimized by introducing systems and solutions that facilitate and provide capabilities for remote witnessing, reducing the time and resources needed to perform the activity. The **REW** notation provides remote audio and video live streaming capabilities that can facilitate efficient communication between vessel and shore, and may be used for multiple purposes typically related to test and inspections activities. This notation also includes optional qualifiers that can be obtained and provide remote witnessing capabilities for systems connected by human machine interface (HMI). This type of solution may be used by different stakeholders, e.g. owner, authorities, equipment manufacturers, warranty surveyors and the Society when found acceptable, see [App.A](#) for more information.

3 Operation and maintenance - hull and structure (OPH)

3.1 General

Hull and structure applicable to main class, service notations and additional class notations will be subjected to periodical surveys to ensure that the structural integrity remains in satisfactory condition and in compliance with the rules, see [DNV-RU-SHIP Pt.7 Ch.1 Sec.1](#).

Operation and maintenance - hull and structure (OPH) enhancements includes solutions that use data as an important element and provide options related to structural integrity management. It also includes solutions that may contribute to increased performance related to the hull and cargo operations.

OPH enhancements may be used to obtain the **OPH** qualifier in **Smart**. For more information, see [App.B](#) for applicable OPH enhancements.

3.2 Structural integrity management

All vessels will be subjected to periodical surveys in accordance with the rules in order to confirm that the hull, structure and hull equipment remain in satisfactory condition and in compliance with approval or accepted standards.

Guidance note:

Hull and structure inspection should be an activity performed by the organization and verified by the inspection activity (periodical surveys) performed by the Society.

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To ensure this, all vessels will be subjected to a survey arrangement, see [Sec.2 \[1.2\]](#) applicable for hull and structure in accordance with requirements of the rules.

Structure and hull inspection may be organized and optimized to limit disruption to vessel operation and reduce the risk as well as the cost associated with the inspection.

Optimal solutions should be determined by identifying potential cost drivers related to the inspection activity and selecting solutions that can mitigate or eliminate them. Cost drivers can vary depending on vessel type and design, service and type of operation or trading, applicable regulatory governance scheme and available inspection resources and tools.

The inspection scope and available options for optimization are usually regulated according to applicable regulatory governance schemes. Some vessels have to comply to a rigid scope such as enhanced survey program (ESP) and as others may apply approaches offering more flexibility such as alternative survey programme (ASP) and risk-based inspections (RBI).

The process of optimizing hull and structure inspection should start at the design phase, evaluating how the design can be optimized to lower the probability of fatigue and other structural damage. This is essential for a reliable operation of the vessel and is not only important to reduce the probability of damage but may also be used in the operational phase to determine an optimized inspection programme with focus on structural elements/areas with high utilization and criticality.

Guidance note:

For design optimisation, it is recommended to apply the systematics described for additional class notations like **FMS**, **CMON**, **Plus**, **CSA**, **CRS** and **RSD** covering direct hydrodynamic and global finite element models (including local finite element models), see [App.B](#) for more information.

Note: **CMON**, **Plus**, **CSA**, **CRS** and **RSD** represent useful and relevant notations, but are not regarded as enhancements that may be used for compliance in **Smart**.

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For structural elements that represent an inspection cost driver and do not allow optimization of the design, or where a design change will involve a significant cost, other solutions should be considered to mitigate or eliminate the cost driver or time consuming inspection activity. This includes:

- risk based inspection (RBI) methodology

- hull and structure condition monitoring (CM) solutions
- use of remote inspection techniques (RIT).

Guidance note:

RBI may be implemented and used as basis for an in-service inspection programme (IIP) for an MOU, see [DNV-RU-OU-0300 Ch.3 Sec.1](#). For other vessel types the use of a RBI programme will in some cases be restricted.

For information about use of RIT, see [DNV-CP-0484 App.A](#) *Special requirements for categories of service suppliers listed in IACS UR Z1*.

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Optimization of hull and structure inspection may also involve an evaluation of inspection resources. This involves use of external resources such as approved service suppliers for NDT, UTM and RIT and also internal resources for visual inspection.

Vessels and the organization may obtain a survey arrangement applicable for hull and structure inspection, allowing the organization (owner) to perform parts of the inspection scope to be performed by owner's qualified personnel.

Guidance note:

For more information and detailed requirements, see [DNV-RU-OU-0300 Ch.2 Sec.4](#) *Survey arrangements-structural - Shared structural inspection*, applicable for MOU's only.

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3.3 Condition monitoring

CM may be applied for any type of hull and structure and used to monitor the condition in terms of fatigue and extreme loading utilization (and potentially corrosion). Fatigue (and corrosion) are regarded as degradation mechanisms and extreme loading as special events. These damage sources require hull and structure inspections to ensure the integrity of the vessel.

Use of CM for hull and structure monitoring may be implemented for a vessel in operation or implemented as part of the design and construction to ensure vessel reliability and limit potential cost drivers in the operational phase. Planning for flexible hull and structure inspection during design may release benefits in operation by utilizing digital twins in combination with inspection findings to extend inspection intervals or achieve improved damage control and transparency. Use of digital twins for the purpose of CM covers indicators, numerical twins, sensors and hybrid (virtual) twins using design models, parametric models and machine learning with the potential of utilizing large data sets (fleet). The output from the digital twins may be used for inspection and maintenance planning directly, or used to develop an in-service inspection programme (IIP) or utilizing remote inspection techniques (RIT). For an IIP it may be relevant with a qualitative and subsequently quantitative RBI assessment to determine which details to monitor and how to monitor them, e.g. NDT or/and close up visual inspections.

A sensor system used as an integrated part of classification should be approved according to requirements described in the additional notation **HMON** for hull monitoring systems.

Guidance note:

For offshore units the digital twin is provided by the additional notation **FMS(NUM)** and **FMS(SENS)**. **CSA**, **CRS** and **RSD** models may also be used as basis for **FMS(NUM)** and **FMS(SENS)**, see [App.B](#) for more information about enhancements that may be used to obtain the **OPH** qualifier.

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4 Energy efficiency and environment (EEN)

4.1 General

The maritime industry is faced with an increasing pressure from regulators, governments and end customers to reduce harmful emissions with outlined strategies to realise zero-emission from shipping in the future.

This calls for the industry to adapt and develop solutions that focus on environmentally friendly shipping and sustainability in the operations.

Energy efficiency and environment (EEN) enhancements are solutions that provide improved energy efficiency and reduced harmful emissions, focusing on technologies that are considered cost-, energy- and emission-efficient for the operation of the vessel. Energy efficiency may address fuel efficiency for engines and generators, novel energy efficient solutions, analytics, hull performance or additional features applied to complement existing non energy efficient technology/solutions.

EEN enhancements also include solutions related to reducing pollution to sea, air or land and may in addition to toxic materials, liquids, gases, garbage, etc., also include aspects of environmental pollution like noise (to sea or air), fumes/smell and vibrations.

Energy efficiency measures will often contribute to environmental improvements. When in place, such capabilities may assist to demonstrate that systems and vessels operate according to established environmental requirements, and may e.g. support environmental compliance reporting schemes.

EEN enhancements may be used to obtain the **EEN** qualifier in **Smart**. For more information, see [App.C](#) for applicable EEN enhancements.

4.2 Performance monitoring solutions

Performance monitoring solutions are processes and means introduced to better understand different aspects of a vessel's or a fleet of vessels' performance often monitored towards set performance indicators.

SECTION 5 SYSTEM QUALIFICATION

1 General

Solutions that are not covered by an existing additional class notation, survey arrangement or defined methodology may be approved and implemented as part of **Smart** as an enhancement. This applies to solutions and technology providing improved capabilities to operation and performance, better energy efficiency or reduction in emission by promoting digitalization and use of data, interoperability and/or connectivity.

Guidance note:

The general process for approval of conventional products is described in [DNV-RU-SHIP Pt.1 Ch.1 Sec.4](#). For more information, see also [DNV-CG-0264 Sec.3 \[3\]](#).

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Novel technology and solutions should be approved according to a system qualification (SQ) process. Any novel technology and alternative solution should represent a relevant function related to the applicable qualifier (**OPM**, **OPH** or **EEN**) normally with an improved effect and/or efficiency compared to an existing or traditional technology or solution. The extent of the approval process will be determined based on the complexity of the applicable technology or solution and if the technology or solution is proven, have limited field history or is new or unproven. The extent of the approval process should be decided based on the criticality of the system or function where the technology and solution are applied.

The SQ process should also determine what type of qualifier the technology and solution may be applied for and how it should be implemented on the vessel according to agreed scope and process.

2 System qualification process

SQ is a process of providing the evidence that a solution or technology will function within specified operational limits with an acceptable level of confidence. SQ may be relevant for novel systems or systems without associated and existing class notations and survey arrangements described in the rules.

The system qualification (SQ) process fits between the process used for approval of conventional technology, and the qualification process used for novel technology as described in [DNV-CG-0264 Sec.3 \[4\]](#). SQ is suited for situations where known technology is applied in new ways or in order to serve new operations. The SQ process should not be used for any critical system or main function described by the rules, see [DNV-RU-SHIP Pt.1 Ch.1 Sec.1 \[2.1.2\]](#).

The SQ process normally includes the following activities and should be agreed in more details in each case:

- SQ basis: establish a qualification basis identifying the technology, its functions, its intended use, as well as the expectations to the technology and the qualification targets.
- Technology assessment: assess the technology by categorizing the degree of novelty to focus the effort where the related uncertainty is most significant and identify the key challenges and uncertainties.
- Threat assessment: assess threats and identify failure modes and their risks.
- SQ plan: develop a plan containing the qualification activities necessary to address the identified risks.
- Execution of the plan: execute the activities specified in the technology qualification plan. Evidence is collected through experience, numerical analyses and tests.
- Performance assessment: assess whether the evidence produced meets the requirements of the SQ basis.

Each step of the process should be documented making the conclusions traceable.

Guidance note:

Models and applications being part of a system qualified by a SQ process should be verified to assure accuracy and performance according to requirements described in [Sec.6](#).

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3 Implementation and testing

Solutions and technology approved following the SQ process will be implemented and tested according to approved test procedure and agreed scope. The implementation should include an agreed survey scope for the Society and applicable maintenance and testing programme for the solution in service.

SECTION 6 MODELS AND APPLICATIONS

1 General

This section provides a high level guidance for verification of models and data-driven applications used as part of applicable enhancements in **Smart** or by other class services when referred to by the rules. The guidance may also be used to document models and data-driven applications part of other technical solutions.

Smart enables use of data as an integrated part of classification systematics and follow up during in-service. Data and information collected from a vessel, fleet of vessels or from other available sources may be applied to solutions enabling real time and well documented decisions. Data may be used for e.g. maintenance optimizations as part of a process to detect, diagnose, predict faults to determine correct maintenance tasks, performance monitoring and utilization calculations with hull monitoring systems.

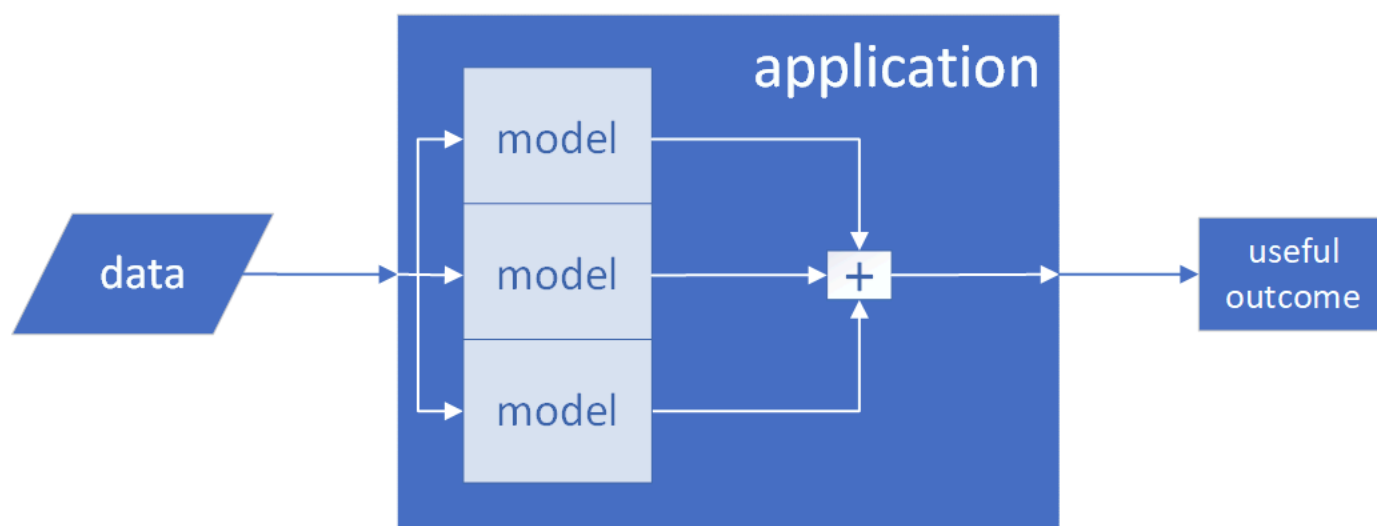


Figure 1 Relation between model and application

A model may be knowledge-based, physics-based or data-driven. A data-driven application, as illustrated in [Figure 1](#), may contain one or more models, and uses the predictions of the model for some business purpose.

Guidance note:

Models and parameters used in an application may sometimes also be referred to as a hybrid model.

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Any type of model and data-driven application used as part of classification shall be approved and any input data shall be quality assured according to the rules and applicable standards. Data should be collected and relayed safely, ensuring a sufficient level of security to prevent and mitigate any potential threat or attack. Data collection infrastructure used for data consumption and sharing should ensure a sufficient level of interoperability to facilitate composite hardware, software, network resources and services required for a sufficient data management.

For more information and requirements for a data collection infrastructure and data management, see [DNV-RU-SHIP Pt.6 Ch.11 Sec.1](#).

2 Verification of models and data-driven applications

Models and data-driven applications may be verified by documenting the development process and the model to a degree sufficient to support the model objective and intended use.

A model or data-driven application should be documented by:

- model objective
- description of which equipment failure mode and/ or degradation it is intended to predict
- model design, including all relevant model(s), sub-model(s) and descriptors/parameters
- baseline
- performance requirements for accuracy, prediction target and probability of detection, see [4]
- development resources' competence and experience
- development process, see [3]
- confidence level, see [5].

3 Development

3.1 General

The development process should be documented according to the type of model:

- knowledge-based model, see [3.2]
- physics-based model, see [3.3]
- data-driven model, see [3.4].

3.2 Knowledge-based model

Knowledge-based models will typically rely on input from subject matter experts (SME) with long time experience (in operation and/or design) with the applicable machine, equipment or system. The following should be documented for a knowledge-based model:

- fault hypotheses with description of associations between failure modes, failure mechanisms, symptoms/ syndromes and descriptors
- definition of alarms or abnormal behaviour according to relevant fault hypotheses.

Guidance note:

It is recommended to apply a failure mode symptoms analysis (FMSA) to document associations between failure modes, failure mechanisms, symptoms/syndromes and descriptors.

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3.3 Physics-based model

A physics-based model encapsulates the physics of a machine, equipment or system based on the design using e.g. first principle equations and finite element methods. This may be used to predict the condition or state using data and information collected from usage/operation. The following should be documented for a physics-based model:

- description of how the condition/state is estimated based on mathematical equation(s) and observed deviations from normal behaviour
- mathematical equations representing the system with an adequate level of decomposition into sub-models elaboration of mathematical equations in a logical sequence
- definition of alarm limits based on design calculations and/or other documented acceptance criteria.

3.4 Data-driven model and application

Data-driven models provide a representation of observed behaviour without the need to have specific understanding of the physics. The following should be documented for a data-driven model and application:

- description of modelling technique used
- data quality (development process)
- representative training data
- representative test data (independent from the training data)
- model or application failure modes
- assessment to verify performance, see [4]
- robustness.

Guidance note:

It is recommended to use systematics described in [DNV-RP-0510 Framework for assurance of data-driven algorithms and models](#) to document a data-driven model or application.

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4 Model performance

Model or data-driven application performance shall be documented in the context of the performance of a baseline.

For knowledge-based and physics-based models the performance should be documented by the accuracy based on level of precision and trueness used to describe the closeness of a measurement to a true value or baseline measurement, see ISO 5725-1. The precision level shall be determined based on several tests sufficient to calculate a standard deviation. Repeatability of tests shall be considered when determining the precision level. Trueness shall be documented by the deviation from an average of measured/calculated value and the real value.

[Figure 2](#) illustrates how precision and trueness are used to describe the accuracy of a measurement or prediction by a model.

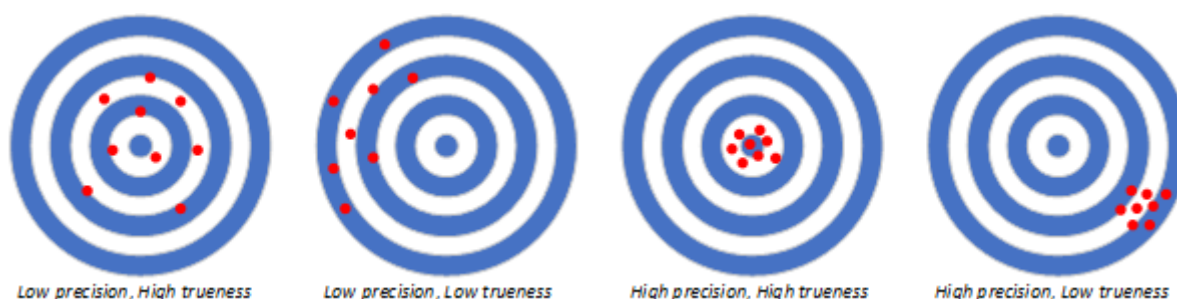


Figure 2 Accuracy based on level of precision and trueness

Guidance note:

High accuracy, see ISO 5725-1, requires both high precision and high trueness.

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Performance of data-driven models and applications should be documented by following the assessment process described in [DNV-RP-0510](#).

5 Confidence level

The confidence level of a model is a figure of merit (e.g. percentage) that indicates the degree of certainty that the prediction (diagnosis/prognosis) is correct. Confidence level should be calculated by using a weighted assessment of the following error sources:

- development (assessment of the process and risks associated with choices made during the development of the model)
- performance evaluation and accuracy, see [4]
- data quality level for descriptors/parameters used, see [DNV-RU-SHIP Pt.6 Ch.11 Sec.1](#)
- system or component design documentation, as applicable
- failure data and characteristics, as applicable.

The calculated confidence level should describe the cumulative effect of error sources on the final certainty of confidence in the accuracy of the prediction.

The confidence level should be compared and evaluated according to the applicable risk level of the applicable function and/or functional failure mode to document and confirm satisfactory confidence level.

Guidance note:

When different types of models are combined in to one (Hybrid) the model should be sub-divided (decomposed) to calculate the confidence level adequately.

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6 Limitations and operating conditions

Models and data-driven applications should be documented with relevant limitations and applicable operating conditions.

APPENDIX A OPERATIONS AND MAINTENANCE - MACHINERY, EQUIPMENT AND SYSTEMS

1 Introduction

The purpose of this appendix is to provide an overview and summary of the enhancements applicable for the **Smart** qualifier **OPM**. Descriptions found in the rules and other references in this appendix are governing, if there are misalignments in the content and descriptions.

2 Enhancements

2.1 Introduction

Enhancements applicable for the **Smart** qualifier **OPM** are split in:

- additional notations, see [2.2]
- survey arrangements, see [2.3].

2.2 Additional notations

2.2.1 Boiler monitoring (automated boiler water maintenance system **BMON(+)**)

The purpose of a boiler monitoring system is to provide an aid to the chief engineer in inspecting the condition of the water/steam side of the vessels boiler(s), and thereby allowing the chief engineer to carry out parts of the boiler survey.

The additional class notation **BMON** sets requirements to a monitoring system which includes documentation and recordings, maintenance, and alarm systems for various parts of the boiler.

The class notation **BMON(+)** indicates that the boiler system is equipped with a type approved automated boiler water maintenance system according to [DNV-CP-0500](#) which is supported by an approved service supplier according to [DNV-CP-0484 App.B](#).

Only systems provided with data from available sensors (e.g. temperatures, pressures, contamination) are regarded as smart.

See [DNV-RU-SHIP Pt.6 Ch.9 Sec.5](#) for design requirements.

See [DNV-RU-SHIP Pt.7 Ch.1 Sec.6](#) for fleet in service requirements.

2.2.2 Data-driven verification (**DDV**)

The additional notation data-driven verification **DDV** is a concept providing methods and systematics for verification of maritime and offshore cyber physical systems (target systems - TS) to be verified based on a data-driven verification method. The principles applied build on ensuring that systems are designed to be verified, and with incorporation of system verify-on-demand capabilities.

DDV may upon request be accepted as additional to, or replace evidence generated from traditional verification activities, that traditionally have relied on deploying a surveyor to the vessel. The intention is that data-driven verification methods may be accepted if the new methods provide the same, or higher, level of assurance as the traditional methods. The standard builds on evolution of current industry practices and seeks to use established and emerging technologies to address the limitations and disadvantages of traditional methods.

The following qualifiers for verification method are included in **DDV**:

- 1) self verifying systems (**SVS**)
- 2) built in test functionality (**BITE**)
- 3) digital survey application (**DSA**)
- 4) algorithm based verification agent (**AVA**)

5) digital twin (**DT**).

Detailed descriptions and requirements for **DDV** are given in [DNV-CG-0557](#) and in [DNV-RU-SHIP Pt.6 Ch.11 Sec.2](#).

2.2.3 Enhanced system verification (ESV)

The additional class notation **ESV** applies to vessels where specified on board systems have undergone enhanced system verification (ESV). A class notation as specified in the rules may be assigned when one or more methods have been used for enhanced system verification. Application of such methods aims to provide additional objective evidence of functionality and quality during normal, abnormal and degraded system status. Application of **ESV** verification provides an additional broader, deeper and earlier verification of the applicable requirements, when compared to main class test activities.

In addition to the generic requirements given in the **ESV** rules, functional requirements and any quality requirements for the stated system(s) shall be specified. In this context all main and additional class requirements applicable for the system(s) in question, apply.

See [DNV-RU-SHIP Pt.6 Ch.5 Sec.13](#) for design requirements.

See [DNV-RU-SHIP Pt.7 Ch.1 Sec.6](#) for fleet in survey requirements.

2.2.4 Integrated software dependent systems (ISDS)

The objective of the class notation **ISDS**, is to reduce the risk for delays in new-build projects and modification projects, as well as for downtime and accidents caused by software in the operation phase.

The **ISDS** notation contains requirements and guidance on the process of design, construction, commissioning and operation of integrated software dependent systems (ISDS). ISDS are integrated systems where the overall behaviour depends on the behaviour of the systems' software components.

The in-service requirements include activities like:

- collection of reliability, availability, maintainability and safety (RAMS) data
- analysis and investigation reports from RAMS incidents/failures
- records of RAMS impact analysis
- security audit reports
- records from validation, verification and testing, as relevant if systems have been changed in operation
- version histories for baselines, requirements trace and configuration records.

See [DNV-OS-D203](#) for design requirements.

See [DNV-RU-OU-0300 Ch.3 Sec.4](#) for fleet in survey requirements.

2.2.5 Integrated navigation system (NAUT(INS+))

The requirements of the **NAUT(INS+)** notation augment the functional requirements of the NAUT-rules and class notations to increase availability and integrity of the information presented on the bridge and thereby improve the situational awareness of the bridge team.

Moreover, requirements for an interconnection of the integrated navigation systems (INS) and the ship-shore communication link is incorporated supporting a secure infrastructure for data exchange with sea traffic management services ashore and facilitating remote condition monitoring and maintenance of the INS.

The scope of **INS+** encompasses a type approved INS, with additional requirements for:

- embedded ethernet
- INS-topology and redundancy arrangement
- enhanced HMI
- interconnection to the ship-shore communication system
- setup and commissioning of the INS.

See [DNV-RU-SHIP Pt.6 Ch.3 Sec.3](#) for design requirements.

See [DNV-RU-SHIP Pt.7 Ch.1 Sec.6](#) for fleet in survey requirements.

2.2.6 Remote witnessing (REW)

REW provides requirements for systems and solutions used for remote witnessing, enabling interaction with technical experts remotely in a safe, flexible and efficient way.

The type of system and solution needed shall consider the criticality and complexity of the inspection and test activity to reflect a sufficient level of insight and transparency when compared to the traditional approach with onboard attendance.

In addition to the general class notation **REW** addressing onboard arrangement for live streaming of activities at bridge, deck, accommodation areas, engine spaces, control rooms, confined spaces, tanks, etc., two (2) different qualifiers may be applied as follows:

- **REW(HMI1)**: indicating that the vessel has functionality for remote witnessing and sharing in real time of onboard control test panels, and that functionality for displaying this in a remote test station is available.
- **REW(HMI2)**: indicating that the vessel has functionality for remote witnessing and sharing in real time of onboard control test panels, and that functionality for displaying this in a remote test station is available, and enabling remote control of the target system HMI interface (operational system control is not required).

See [DNV-RU-SHIP Pt.6 Ch.11 Sec.3](#) for design requirements.

2.3 Survey arrangements

2.3.1 Drilling equipment condition based maintenance (DCBM)

For units with an approved drilling equipment condition based maintenance survey arrangement (DCBM), condition based maintenance may replace the relevant survey requirements of [DNV-RU-OU-0300 Ch.3 Sec.4 \[6.2\]](#) and [DNV-RU-OU-0300 Ch.3 Sec.4 \[6.3\]](#), for applicable equipment and failure modes.

The arrangement shall be approved in accordance with machinery CBM, see [DNV-RU-OU-0300 Ch.3 Sec.2 \[5\]](#).

See [DNV-RU-OU-0300 Ch.2 Sec.3](#) for fleet in survey requirements.

2.3.2 Electrical system integrity management (ESIM)

Electrical systems integrity management (ESIM) may be applied to main class electrical installations as covered by [DNV-RU-OU-0300 Ch.3 Sec.1](#).

ESIM is offered as an integral part of classification compliance for main class electrical equipment and systems through the alignment and integration of classification requirements with an approved and implemented inspection and test programme.

Survey arrangement machinery planned maintenance system (MPMS), see [\[2.3.5\]](#), is required both for implementing and maintaining survey arrangement ESIM.

ESIM allows for parts of the inspection and testing programme to be performed by owner's qualified personnel. An inspection and testing programme shall be approved and implemented.

When an ESIM has been implemented successfully on a unit, this will give an alternative to the traditional survey and testing of electrical installations described in the rules.

See [DNV-RU-OU-0300 Ch.2 Sec.2](#) for fleet in survey requirements.

2.3.3 Machinery condition based maintenance (MCBM)

Machinery condition based maintenance (machinery CBM) is a survey arrangement applicable for selected machinery components part of the vessel machinery list. Maintenance tasks are established and planned based on one or more condition monitoring (CM) methods, diagnostics and prognostics. The survey arrangement may be applied for:

- selected functional failure mode(s) or include
- all functional failure modes applicable for the complete machinery component.

A CM method may be based on the following descriptors (input):

- standard CM measuring technique(s), see ISO 18436

- failure modelling technique, see [Sec.6](#)
- parameters (non-manipulated reflecting the operating conditions of the machine)
- testing and inspection.

Machinery CBM requires use of an approved service supplier for condition monitoring, diagnostics and prognostics of selected machinery for applicable failure modes. When machinery CBM has been implemented successfully on a vessel, this will give an alternative to the traditional predetermined maintenance given by fixed intervals. With an approved machinery CBM, the required maintenance will be performed according to the condition of the machinery.

See [DNV-RU-OU-0300 Ch.2 Sec.2](#) for fleet in survey requirements for MOUs.

See [DNV-RU-SHIP Pt.7 Ch.1 Sec.7](#) for fleet in survey requirements for ships.

See also [DNV-CG-0052](#) for further details.

2.3.4 Mooring integrity management position mooring system (MIM)

In case a mooring integrity management (MIM) programme is developed for a vessel and this has been subject to class review and approval, this programme shall be followed instead of the listed requirements to annual and complete surveys as described in the **POSMOOR** rules.

See [DNV-RU-OU-0300 Ch.3 Sec.4](#) for fleet in survey requirements.

2.3.5 Machinery planned maintenance system (MPMS) applying machinery maintenance connect (MMC) analytics

Machinery maintenance connect (MMC) is an analytical method and dashboard used to carry out the machinery planned maintenance survey arrangement (MPMS).

MMC is based on data sent from an approved and implemented planned maintenance system (PMS) on board which covers all class-relevant component surveys related to machinery. MMC features a dashboard which displays information about machinery maintenance status for each individual vessel and it allows the Society to perform the MPMS surveys for the entire fleet at once. It also allows the vessel owner to follow up on the progress of on-board maintenance.

Guidance note:

Data integration into MMC is not a complicated task, as the major PMS software providers are already fully integrated in this application. It is also possible to get the dashboard into the company's PMS software.

---e-n-d---o-f---g-u-i-d-a-n-c-e---n-o-t-e---

By applying MMC to a group or fleet of vessels with an approved MPMS it allows organizations (owners) to get a better overview, draw more learning and better direct their maintenance system across the fleet.

See [DNV-RU-SHIP Pt.7 Ch.1 Sec.7](#) for fleet in survey requirements.

2.3.6 Marine riser management (MRM)

Marine riser management (MRM) is applicable for vessels with **DRILL** notation and applies to elements of the marine riser system, e.g. riser joints, telescopic joints and ball/flex joints, which are part of the drilling items registered to the vessel.

MRM is an alternative to a regular inspection, maintenance, and repair approach following a fixed five (5) years cycle. MRM shall be based on a continuous approach to ensure that all failure modes and degradations impacting the integrity of the riser(s) are identified, quantified and actions taken to ensure that acceptable level of safety is achieved.

MRM shall include the methodology for both usage and conditions logging.

See [DNV-RU-OU-0300 Ch.3 Sec.4 \[6.6\]](#) for fleet in survey requirements.

2.3.7 Planned maintenance system, reliability centered (PMS RCM)

The requirements for machinery planned maintenance system - reliability centred (MPMS RCM) are based on a proactive maintenance approach with an objective to enhance maintenance effectiveness and focus on the owner's ability to provide required maintenance activities when needed.

A maintenance management approval is performed to ensure that the owner of the vessel is able to provide satisfactory maintenance support within the organization. The approval shall include a review of documentation covering:

- maintenance management (process) and support performance
- resources, competence and responsibilities
- process for monitoring performance and continuous improvement
- prioritizing, supervision and verification of work
- governing documentation (procedures) and working processes
- spare part management.

See [DNV-RU-OU-0300 Ch.2 Sec.2](#) for fleet in survey requirements.

2.3.8 Thruster condition monitoring (Thruster CM)

The survey arrangement for thruster CM, based on vibration measurements and lubrication- and hydraulic oil sampling, may replace the complete survey of thrusters. It is intended to monitor the condition of the mechanical power transmission train (i.e. bearings and gears) and may also be implemented for podded type thrusters, i.e. thrusters equipped with integrated electric drive motor, instead of gear transmission, to monitor shaft bearings and sealing condition.

Depending on the function of the monitored thruster, the oil sampling intervals shall be adjusted to a minimum:

- monthly oil sampling and - analysis of thrusters for propulsion
- quarterly oil sampling and - analysis of thrusters for position keeping.

See [DNV-CG-0052](#) for design requirements.

See [DNV-RU-SHIP Pt.7 Ch.1 Sec.5](#) for fleet in survey requirements.

APPENDIX B OPERATION AND MAINTENANCE - HULL AND STRUCTURE

1 Introduction

The purpose of this appendix is to provide an overview and summary of the enhancements applicable for the **Smart** qualifier **OPH**. Descriptions found in the rules and other references in this appendix are governing, if there are misalignments in the content and descriptions.

2 Enhancements

2.1 General

Enhancements applicable for the **Smart** qualifier **OPH** are split in:

- additional notations, see [2.2]
- survey arrangements, see [2.3].

2.2 Additional notations

2.2.1 Alternative survey programme, motion reference unit installed (ASP(MRU))

The purpose of **ASP** is to give alternative requirements to enable liquefied gas storage units (e.g. FSU and FSRUs) a more flexible scheduling of internal cargo tank survey and remain at the same safety level as vessels doing full internal cargo tank survey every five years for all cargo tanks.

Only systems utilizing motion reference unit (MRU) and other sensors (e.g. gas concentration, N₂ consumption, temperature) are regarded as smart. Additional equipment may be required to be installed for ASP, e.g. pumps for redundancy. These vessels will have the class notation **ASP(MRU)**.

See DNV-RU-SHIP Pt.6 Ch.9 Sec.8 for design requirements.

See DNV-RU-SHIP Pt.7 Ch.1 Sec.6 for fleet in survey requirements.

2.2.2 Fatigue methodology specification (FMS)

FMS is a design notation for fatigue methodology. To enable re-use of the model(s) in operation the additional qualifiers **NUM** or **SENS** may be assigned to units where structural condition monitoring during operation are carried out according to requirements in the rules.

The objective of the qualifiers **NUM** and **SENS** is to extend the value of the **FMS** notation and make the design models operational for structural condition monitoring, which can be used for decision support and improved inspection and maintenance planning.

The qualifiers **NUM** and **SENS** are applicable for all types of structures and can be used to monitor fatigue and strength utilisation in operation. This is an expansion from the **FMS(Y)** notation that covers ship shaped units. For other unit types than ship-shaped, considerations will be made to ensure that the FE model(s) are suitable for the intended purpose.

FMS(NUM) re-uses the structural models created in the design phase as part of **FMS(Y)** notation. In service the design assumption of the wave environment and loading conditions are replaced by the encountered wave conditions from available weather data and loading conditions during operation to predict the fatigue damage. The owner shall provide the Society with automatic identification system (AIS) data including heading and speed (when relevant).

FMS(SENS) re-uses the structural models created in the design phase as part of the **FMS(Y)** notation. The scope includes planning and instrumentation of the unit with a hull stress monitoring system according to **HMON** requirements. Placement and number of sensors shall be according to the rules. If the owner wants to use a previously installed hull monitoring system that is not part of the **HMON** class notation, such systems shall be evaluated for suitability in accordance with **HMON** requirements.

See [DNV-RU-OU-0300 Ch.3 Sec.4](#) for fleet in survey requirements.

2.2.3 Hull monitoring systems (HMON)

A vessel equipped with a hull monitoring system and assigned the additional class notation **HMON** is equipped with a sensor monitoring system for measuring different quantities relevant during operation of a vessel. The measured quantity, as described in the **HMON** rules in [DNV-RU-SHIP Pt.6 Ch.9 Sec.3](#), shall be displayed onboard for decision support.

Measured sensor data shall be collected for onshore decision support.

The hull monitoring system may include sensors for several purposes, e.g.:

- global structural response of the hull
- local structural response of the hull
- ice response monitoring
- environmental monitoring
- seamanship
- navigation
- comfort and vibration
- rotating machinery
- propulsion system
- rudder performance
- motions and accelerations
- parametric roll
- pressures, sloshing and slamming.

The monitoring system is intended as an aid to the master's judgement onboard and not as a substitute for it.

The use as an enhancement for the **Smart** qualifier **OPH** is applicable for **HMON(B)** and **HMON(D)**.

See [DNV-RU-SHIP Pt.6 Ch.9 Sec.3](#) for design requirements.

See [DNV-RU-SHIP Pt.7 Ch.1 Sec.6](#) for fleet in survey requirements.

2.2.4 Loading computer systems for stability and longitudinal strength - damage control (LCS(DC))

LCS(DC) puts requirements to a loading computer system to assist the master as a decision aid under damage conditions.

The scope for the additional class notation **LCS** concerns an on-board computer system running a combination of software modules for stability and strength verification. The term loading computer systems applied in the rules also covers the terms loading instruments and on-board stability computers. The rules do not cover basic (system) software such as device drivers necessary for the hardware to support the application software. The qualifier **DC** added to the notation (i.e. **LCS(DC)**) concerns damage control software developed to assist the master as a decision aid, receiving input from flooding and filling detectors installed in the vessel's compartments.

See [DNV-RU-SHIP Pt.6 Ch.4 Sec.6](#) for design requirements.

2.2.5 Remote witnessing (REW)

See [App.A \[2.2.6\]](#) for more information.

2.2.6 Route specific container stowage based on weather forecast (RSCS+)

The additional class notation **RSCS+** provides requirements for load assumptions for lashing calculations, applicable to ships with route specific reductions in transverse design. The applicable predefined or individual routes, as well as route tolerances are described in the rules. A customer may specify individual routes based on a sequence of port calls or waypoints applied in the respective **RSCS+** application available on www.veracity.com. Containers are stowed on-board the ship in accordance with an approved route

specific addendum to the container securing arrangement plan. The route description shall not hazard safe navigation, which means that adverse weather conditions should be avoided as far as possible.

The additional class notation **RSCS+** applies to container ships with route specific container stowage. The requirements shall be regarded as supplementary to main class rules, and are optional for ships with notation **Container ship** or **Container**. Each specific route reduction factor shall be calculated by the **RSCS+** application. Further weather dependent acceleration reductions can be applied for short voyages as described in the rules.

See [DNV-RU-SHIP Pt.6 Ch.4 Sec.10](#) for design requirements.

See [DNV-RU-SHIP Pt.7 Ch.1 Sec.2](#) and [DNV-RU-SHIP Pt.7 Ch.1 Sec.4](#) for fleet in survey requirements.

2.3 Survey arrangements

2.3.1 Crane - condition assessment

Crane or **Offshore crane** applies to any type of offshore crane intended for cargo handling outside the unit while at sea and to any type of platform crane intended for cargo handling on the unit.

As an alternative to the survey scope described in the rules, a condition assessment of the crane may be performed. The condition assessment shall be based on design working period calculations. Based on the condition assessment, a repair, replacement and inspection plan shall be established and implemented.

As part of the condition monitoring the following steps shall be completed:

- identification of design and installation data
- recording necessary in operation data
- calculating the design working period
- performing special assessment
- reporting all necessary information, calculations and assessment
- establishing a repair, replacement and inspection programme.

The above steps shall be performed according to ISO 12482.

See [DNV-RU-OU-0300 Ch.3 Sec.4](#) for fleet in survey requirements.

APPENDIX C ENERGY EFFICIENCY AND ENVIRONMENT

1 Introduction

The purpose of this appendix is to provide an overview and summary of the enhancements applicable for the **Smart** qualifier **EEN**. Descriptions found in the rules and other references in this appendix are governing, if there are misalignments in the content and descriptions.

Enhancements applicable for the **Smart** qualifier **EEN** are split in:

- additional notations, see [2]
- performance monitoring solutions, see [3].

2 Additional notations

2.1 Ballast water management - ballast water treatment (**BWM(T)**)

The International Convention for the Control and Management of Ships' Ballast Water and Sediments (BWM) (commonly referred to as the convention), entered into force on 2017-09-08. The convention provides regulations to prevent transfer and introduction of harmful aquatic organisms and pathogens via ballast water. The requirements for the voluntary class notation **BWM** are in compliance with the convention.

The class notation **BWM(T)** specifies requirements for ballast water treatment system installations. Relevant procedural requirements are included for ships using ballast water in international trade and for which the convention is applicable.

Ballast water management treatment is mandatory for all new (2017) international sea going vessels under the convention and will be made mandatory for existing vessels by 2024. Some vessels may be exempted from complying with the convention, according to certain conditions and the appropriate permission being given by a flag state.

See [DNV-RU-SHIP Pt.6 Ch.7 Sec.1](#) for design requirements.

See [DNV-RU-SHIP Pt.7 Ch.1 Sec.6](#) for fleet in survey requirements.

2.2 Battery(Power)

The additional class notation **Battery(Power)** facilitates the use of electrical energy storage (EES) installations on electric and hybrid vessel and sets requirement for safety and availability of EES installations onboard vessels.

Battery(Power) applies for vessels where the EES power is used for electrical propulsion of the vessel.

A battery management system (BMS), a capacitor management system (CMS), as relevant for an electrochemical capacitor system, as well as an energy management system (EMS) shall be installed and certified as a part of the system.

See [DNV-RU-SHIP Pt.6 Ch.2 Sec.1](#) for design requirements.

See [DNV-RU-SHIP Pt.7 Ch.1 Sec.2](#) and [DNV-RU-SHIP Pt.7 Ch.1 Sec.4](#) for fleet in survey requirements.

2.3 Environmental class (**Clean(Design)**)

The additional class notations **Clean(Design)** and **Clean(Design, Tier III)** set requirements for ship's design, operation and equipment reducing the environmental impact from emissions to air, discharges to sea, and deliveries to shore from vessels. The requirements for these additional class notations are in compliance with, or more extensive than those found in international standards currently in force.

The scope of the additional class notations **Clean(Design)** and **Clean(Design, Tier III)** is to attain a vessel with controlled environmental standards of design and performance. Compliance with the rules shall be

verified through inspection, measurements and sampling of defined environmental parameters in accordance with the requirements of the rules and in compliance with identified standards and guidelines.

See [DNV-RU-SHIP Pt.6 Ch.7 Sec.2](#) for design requirements.

See [DNV-RU-SHIP Pt.7 Ch.1 Sec.6](#) for fleet in survey requirements.

2.4 Comfort class (noise and vibration) (**COMF(V-3)**)

The additional class notation **COMF(V-3)** is applicable for all ship types. The class notation **COMF(V-3)** indicates that comfort conditions that may adversely influence human performance on board ships have been taken into consideration, and that the vessel satisfies acceptable levels of noise and vibration and indoor climate on board ships. The qualifier **V-crn** indicates to what level the requirements related to noise and vibration have been achieved.

The additional class notation **COMF(V-3)** includes requirements for noise levels, vibration levels and the performance of the on board HVAC system, for passenger ships, cargo ships, yachts and high-speed crafts. The areas included are cabins, public spaces, wheelhouse, radio rooms, offices, control rooms, hospital and recreation rooms.

Compliance with the rules shall be verified through measurements of defined environmental parameters. Possible influence of vibration on machinery, structure or other systems is considered in other relevant society rules. The least strict noise level and sound insulation requirements (crn = 3) for cargo ships correspond to the requirements in IMO Resolution MSC.337(91) Code on noise levels on board ships. In order to achieve the class notation, the vibration criteria's in the rules shall also be satisfied.

See [DNV-RU-SHIP Pt.6 Ch.8 Sec.1](#) for design requirements.

2.5 Emission control areas (**ECA**)

The scope of the additional class notation **ECA** shall verify that the vessel is arranged to enable vessel's machinery components (main propulsion plant, power generation plant and steam/thermal oil plant, inert gas plant, etc.) to change between residual oil and marine distillate fuel and operate for longer periods on marine distillate fuels with very low viscosity and very low sulphur content.

The additional class notation **ECA** applies to vessels arranged and equipped as required by the rules. Vessels with the additional class notation **ECA** may be appended by one of the mandatory qualifiers

SOx-A or **SOx-P** where:

- **ECA** denotes that the vessel is adapted to operate within emission control areas.
- **SOx** denotes that the vessel is adapted to comply with SOx regulations within emission control areas as per Annex VI of MARPOL 73/78 and can operate specific machinery components on marine distillate fuels with very low viscosity and sulphur content for a minimum of four (4) operating days reflecting the consumption specified.
- **A** denotes that the vessel is designed to operate all machinery components on marine distillate fuel.
- **P** denotes that the vessel is designed to only operate machinery components used in port on marine distillate fuel.

See [DNV-RU-SHIP Pt.6 Ch.7 Sec.3](#) for design requirements.

2.6 Emission reduction (**ER**)

The additional class notation **ER**, emission reduction, provides criteria for the safe and environmentally friendly arrangement and installation of machinery for NOx and SOx exhaust gas emission reduction.

The scope includes requirements for the design and arrangement of EGCS, SCR and EGR systems, including the following:

- piping systems conveying wash water and/or treatment fluids
- exhaust arrangements and components
- control, monitoring and safety systems and components

- manufacture, workmanship and testing.

Different qualifiers apply.

See [DNV-RU-SHIP Pt.6 Ch.7 Sec.7](#) for design requirements.

See [DNV-RU-SHIP Pt.7 Ch.1 Sec.2](#) and [DNV-RU-SHIP Pt.7 Ch.1 Sec.4](#) for fleet in survey requirements.

2.7 Fuel cell installations (**FC**)

The scope for **FC** includes requirements for the design and arrangement of fuel cell power installations and the spaces containing such installations. It covers all aspects of the installation from primary fuel supply up to, and including, the exhaust gas system. Further the following is covered:

- reformers used to convert liquid or gaseous primary fuels to reformed hydrogen rich gas
- control, monitoring and safety systems
- manufacture, workmanship and testing.

The use of fuel cells is currently not covered by international conventions, hence such installations will require additional acceptance by the flag authorities.

Vessels complying with the requirements given in the rules will be assigned the additional class notation **FC**, with qualifiers as follows:

- **FC(Power)**: fuel cell power installations intended for supply of essential, important or emergency services
- **FC(Safety)**: fuel cell power installations not supplying essential, important or emergency services.

See [DNV-RU-SHIP Pt.6 Ch.2 Sec.3](#) for design requirements.

See [DNV-RU-SHIP Pt.7 Ch.1 Sec.2](#), [DNV-RU-SHIP Pt.7 Ch.1 Sec.3](#) and [DNV-RU-SHIP Pt.7 Ch.1 Sec.4](#) for fleet in survey requirements.

2.8 External airborne noise emission (**Quiet**)

Airborne noise pollution is an important part of the environmental impact of ships, where it may affect living, working and leisure areas near ports. The overall external airborne noise emission radiated from a vessel can be traced back to several individual sources. Based on experience, the most relevant external sources of noise are funnel outlets of the engines, the openings of engine room ventilation inlets and outlets and the openings of the cargo holds- and accommodation ventilation and air-conditioning. There is an increasing need to be able to document the actual noise emitted from a vessel berthed in a port, in idle or under working conditions to satisfy port authority and/or in order to qualify for reduced port fees.

Quiet does not imply permanent measurements. Only permanent instrumentation to measure noise emissions are regarded as an enhancement with respect to the **Smart** notation.

See [DNV-RU-SHIP Pt.6 Ch.7 Sec.8](#) for design requirements.

2.9 Shore power

The objective of the additional class notation **Shore power** is to provide requirements for a transfer of power utilizing an electrical shore connection while in port.

The additional class notation **Shore power** provides requirements for the design of electrical shore connections, the ship side installation of necessary equipment and the verification of the installations.

The system design comprises the following aspects:

- system functionality of the electrical shore connection as a total system. In addition, requirements to circuit breakers, earthing switches and protective functions are given
- control systems and control system interface between the shore and the vessel
- ship side electrical equipment and installations.

See [DNV-RU-SHIP Pt.6 Ch.7 Sec.5](#) for design requirements.

2.10 Underwater noise emission (**Silent**)

For some vessels, hydro-acoustic transducers are important tools for operating the vessel efficiently. For all vessels, noise emission is considered an environmental disturbance. The additional class notation **Silent** ensures that vessels with a need for a controlled underwater noise emission can be designed and tested towards technically realistic requirements.

Silent does not imply permanent measurements. Only permanent instrumentation to measure noise emissions are regarded as an enhancement with respect to the **Smart** notation.

See [DNV-RU-SHIP Pt.6 Ch.7 Sec.6](#) for design requirements.

2.11 Wind assisted propulsion system (**WAPS**)

Wind assisted propulsion systems (WAPS) like rotor sails or wing rigs can be installed on different kinds of ships. The purpose of such WAPS is the contribution to reduce the consumption of conventional fuels. WAPS shall not replace conventional propulsion means, but assist in favourable operating conditions.

The additional class notation **WAPS** may be applied to all ship types with a wind assisted propulsion system installed. The **WAPS** installation shall be considered as an auxiliary propulsion system, thus the ship shall be fully operable with main propulsion only.

See [DNV-RU-SHIP Pt.6 Ch.2 Sec.12](#) for design requirements.

3 Performance monitoring solutions

3.1 General

Performance monitoring solutions refer to electronic/digital means or analytical engines applied to monitor the performance of key activities, parameters, targets or requirements related to compliance performance.

The solutions may support users in demonstrating that systems and vessels operate according to established regulatory or operator requirements.

3.2 Approved electronic record book(s) (ERB) in use

With the maritime industry getting more digital, electronic recording books (ERBs) are being gradually applied. Recent MARPOL and NOx Technical Code amendments now also enable the use of electronic record books in lieu of hard copy record books. Electronic recording may benefit the retention of records by companies/ships and will reduce the administrative burden associated with paperwork.

The MARPOL and NOx Technical Code amendments entered into force on 1 October 2020. The regulations enable the electronic recording of the following record books:

- Oil Record Book, part I (MARPOL Annex I)
- Oil Record Book, part II (MARPOL Annex I)
- Cargo Record Book (MARPOL Annex II)
- Garbage Record Book, part I and II (MARPOL Annex V)
- Ozone-depleting Substances Record Book (MARPOL Annex VI)
- Recording of the tier and on/off status of marine diesel engines (MARPOL Annex VI)
- Record of Fuel Oil Changeover (MARPOL Annex VI)
- Record Book of Engine Parameters (NOx Technical Code).

Before replacing a hardcopy record book, the electronic record book (ERB) will be subject to approval by the flag state. It is also recommended to ensure that relevant port states accept ERBs.

For requirements to ERBs, see [DNV-CP-0569](#) - *Electronic record book* -, ISO 21745 - *Electronic record books for ships - technical specification and operational requirements*, and MEPC.312(74) - *Guidelines for the use of electronic record books under MARPOL*.

3.3 IMO DCS - IMO Data Collection System on fuel consumption, and Emissions Insights

Both IMO and EU have clear ambitions to reduce greenhouse gas emissions from ships. The IMO DCS - IMO Data Collection System on fuel consumption started their data collection on 1st of January 2019. It is applicable to ships above 5000 gross tonnage (GT) trading globally.

The DNV DCS service offers a fully digital reporting approach, with online tools to allow checking the quality of reported voyage and emission data through the reporting period. The first step towards significantly reducing the greenhouse gas emissions from ships is to monitor the fuel consumption and CO₂-emissions. The DNV DCS service is going beyond DCS compliance by providing *Emissions Insights*, an insight dashboard that gives insights to the reported emission data. The dashboard enables managers to have much better control over the emission status of their vessels, and gives support for decisions to ensure compliance, competitiveness and the saving of costs.

For reporting and performance requirements, see DNV IMO DCS landing page (<https://www.dnv.com/maritime/insights/topics/EU-MRV-and-IMO-DCS/index.html>).

For details on IMO DCS, see MEPC.278(70) (<http://www.imo.org/en/OurWork/Environment/PollutionPrevention/AirPollution/Documents/278%2870%29.pdf>).

3.4 EU MRV - EU Monitoring, Reporting and Verification of CO₂ emissions with the DNV automatic reporting tool

Both, IMO and EU have clear ambitions to reduce greenhouse gases emissions from ships. DNV supports the customers with two similar, albeit separate, introduced regimes.

One of the regimes is an automatic reporting tool that has been introduced by DNV for the EU MRV - EU Monitoring, Reporting and Verification of CO₂ emissions (data collection started 1 January 2018).

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This is a new document.

About DNV

DNV is the independent expert in risk management and assurance, operating in more than 100 countries. Through its broad experience and deep expertise DNV advances safety and sustainable performance, sets industry benchmarks, and inspires and invents solutions.

Whether assessing a new ship design, optimizing the performance of a wind farm, analyzing sensor data from a gas pipeline or certifying a food company's supply chain, DNV enables its customers and their stakeholders to make critical decisions with confidence.

Driven by its purpose, to safeguard life, property, and the environment, DNV helps tackle the challenges and global transformations facing its customers and the world today and is a trusted voice for many of the world's most successful and forward-thinking companies.

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